Wealth Inequality and growth

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Introduction

- Last week: Discussed growth through factor (capital) accumulation
  - Generally, seen through the lens of a represenative agent
- This week we discuss the nexus between growth and wealth distribution
  - Later in the course we will discuss how growth and wage inequality intersect
- We begin by asking how distribution may matter to growth; subsequently we reverse the chain of causality.

Roadmap

1. **From distribution to growth (Alesina-Rodrik)**
    1. Overview of model
    2. Household behavior, Firm behavior and the macroeconomy
    3. Political Economy: Selecting taxes
    4. Predictions, Evidence, and remarks on literature
2. From growth to distribution (Krusell and Smith)
    1. Data
    2. Piketty's first theory
    3. Summary, critique and remarks on literature

From Distribution to Growth

Overview of the model

- The basic model is an AK model, where growth is fuelled by factor accumulation through productive government investments (B&S Ch. 4)
- New element: Individual's relative endowment of capital varies. There is inequality.
- Distribution of wealth and income matters to long-run growth. Why? Because the tax rate on capital income is endogenously determined through majority voting. Adresses the question: If differences in taxes generate growth differences across countries, then why do taxes vary?
    - Ceteris paribus: More inequality means higher taxes, which means lower growth. Reduced form: Inequality is bad for growth.
- Let's look at the details ..

From Distribution to Growth

Household behavior

- The individual household, indexed by i, faces the following problem (time index suppressed)

$$\max_{\{c\}_0^\infty} \int_0^\infty \ln\left(c_i\right) e^{-\rho t} dt$$

s.t.

$$c_t > 0$$

$$\dot{k}_i = wl_i + \left(r - \tau\right) k_i - c_i, \;\; k_i \text{ given}$$

$$\lim_{t \to \infty} k_i e^{-(r-\tau)t} = 0$$

- Note that $\sum_i l_i = L$ (and *constant*), and $\sum_i k_i = K$.

From Distribution to Growth

Household behavior

- Usual steps (see slides from lecture 1) leads to the Consumption Euler

$$\frac{\dot{c}_i}{c_i} = r - \tau - \rho$$

- Note that with log preferences we have that the elasticity of intertemporal substitution in consumption is 1 ($\lim_{\theta \to 1} \frac{c^{1-\theta}-1}{1-\theta} = \ln(c)$).
- Key thing to notice: If all households have identical preferences, and face the same real rate of interest and tax rate, they will have their consumption grow *at the same rate*

$$\frac{\dot{c}_i}{c_i} = \frac{\dot{c}_j}{c_j} \text{ for all } i \neq j.$$

- Thus, per capita consumption grows at the rate $r - \tau - \rho$ as well

$$\frac{\dot{c}_i}{c_i} = \frac{\dot{c}_j}{c_j} = \frac{\dot{c}}{c}.$$

From Distribution to Growth

Firms

- The representative firm maximizes

$$\max_{l,k} Y - wL - rK$$

where

$$Y = AK^{\alpha} (GL)^{1-\alpha}$$

- We get the usual first order conditions

$$r = \alpha \frac{Y}{K}, \ w = (1-\alpha) \frac{Y}{L}$$

From Distribution to Growth

Government

- We will assume that the government uses all revenue on productive government investments and thus balances the budget

$$G = \tau K.$$

- The tax rate τ is assumed constant. But its level will in the end be endogenous, and decided upon through majority voting.
- Using the balanced budget restriction in the production function of the represenative firm

$$Y = AK^{\alpha}(GL)^{1-\alpha} = AK^{\alpha}(\tau KL)^{1-\alpha} \equiv \bar{A}(\tau)K$$

with $\bar{A}(\tau) = A(\tau L)^{1-\alpha}$, $A'_{\tau} > 0, A''_{\tau\tau} < 0$. Hence, endogenous growth will arise: Constant returns to reproducible factor (cf Lecture 1)

From Distribution to Growth

Macro dynamics

- Aggregate dynamics. With an AK-model in hand we know: (i) NO transitional dynamics, (ii) Balanced growth at all points in time (cf lecture 1). In particular

$$\frac{\dot{Y}}{Y} = \frac{\dot{K}}{K} = \frac{\dot{c}}{c}.$$

- **This also implies** $\dot{c}_i / c_i = \dot{k}_i / k = \dot{k}_j / k_j = \dot{c}_j / c_j$ for all $i \neq j$!
- We already know how fast per capita consumption grow via the Consumption Euler. All we need to do is to work out what r is. From firm FOC

$$r = \alpha \frac{Y}{K} \underbrace{=}_{Y = A(\tau)K} \alpha \bar{A}(\tau).$$

Hence output grows at the rate (which depends on the level of taxation; to be determined).

$$\gamma = \alpha \bar{A}(\tau) - \tau - \rho.$$

From Distribution to Growth

Macro dynamics

- Just as in the model in the exercises:

$$\frac{\partial \gamma}{\partial \tau} - 1 \gtreqless 0 \Leftrightarrow \tau \lesseqgtr \tau^*$$

- where

$$\tau^* = \left[\alpha (1-\alpha) A \tau L^{1-\alpha}\right]^{1/\alpha}$$

  $\tau^*$is the growth maximizing tax rate. What explains this *non-linear* link?

- Which level will prevail, if the agents vote over taxes? That is the next question...

From Distribution to Growth

Political Economy: Selecting the tax rate

- The problem for the individual agent is to choose the tax rate he or she prefers. That is, which maximizes discounted utility over an infinte horison, $\int_0^\infty \ln c_i(t) e^{-\rho t} dt$.
- We assume taxes are voted for once, and kept constant (can be relaxed, cf Strulik and Lindner, 2004). The maximization problem can thus be written as a simpler static optimization problem, as we now show.
- Observe that since we are on a balanced growth path, the level of consumption fulfills

$$\gamma = \frac{\dot{k}_i}{k_i} = \underbrace{(1-\alpha)\bar{A}(\tau)\frac{K/L}{k_i/l_i}}_{wl_i/k_i} + (r-\tau) - \frac{c_i}{k_i} \overset{\text{use } \gamma = r-\tau-\rho}{\Rightarrow}$$

$$c_i = (1-\alpha)\bar{A}(\tau)k + \rho k_i = [(1-\alpha)\bar{A}(\tau)\sigma_i + \rho] k_{i0} e^{\gamma t}$$

where $\sigma_i \equiv (K/L)/(k_i/l_i)$, which is constant over time due to balanced growth.

- We will use σ_i as an index of each agents endowment

$$\sigma_i = \frac{K/L}{k_i/l_i}.$$

- Note that a "perfect capitalist" (no labor $l_i = 0$) have $\sigma_i = 0$. Whereas $\sigma_i = \infty$ is an individual without capital.
- Since we have the entire path for c_i as a function of taxes, we can just solve the integral (employ integration by parts):

$$\begin{aligned}\int_0^\infty \ln\left(c_i(0)\, e^{\gamma t}\right) e^{-\rho t} dt &= \ln c_{i0} \int_0^\infty e^{-\rho t} dt + \gamma \int_0^\infty t e^{-\rho t} dt \\ &= \frac{1}{\rho}\left(\ln c_i(0) + \frac{\gamma}{\rho}\right)\end{aligned}$$

From Distribution to Growth

Political Economy: Selecting the tax rate

- The problem, then, boils down to

$$\max_{\tau} \frac{1}{\rho}\left(\ln c_i(0) + \frac{\gamma}{\rho}\right),$$
$$s.t.\ \gamma = \alpha\bar{A}(\tau) - \tau - \rho,\ c_i(0) = \left[(1-\alpha)\bar{A}(\tau)\sigma_i + \rho\right]k_{i0}$$

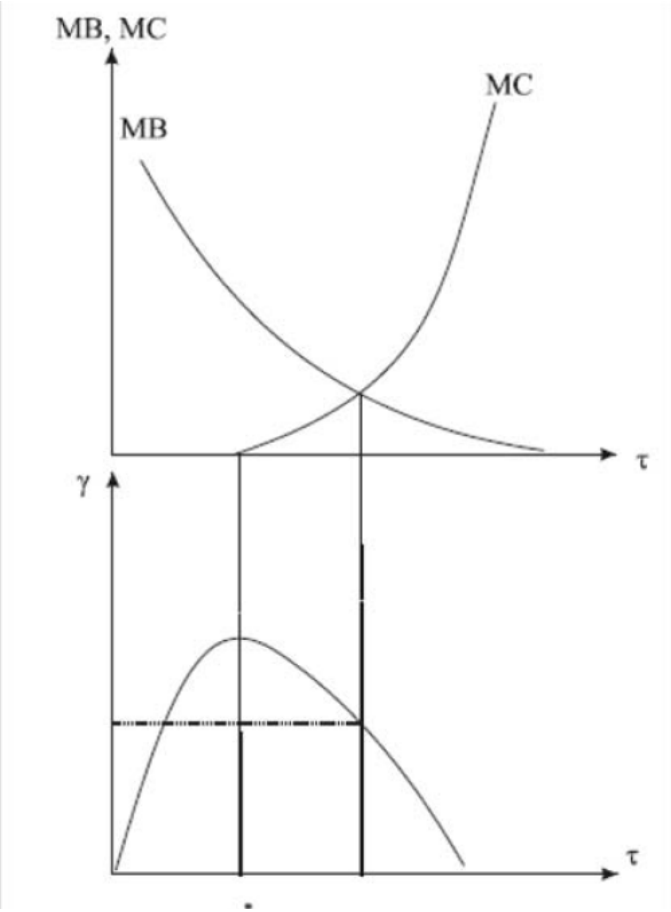
- The first order condition:

$$\underbrace{\frac{\partial c_i(0)}{\partial \tau}\frac{1}{c_{i0}}}_{\mathrm{MB}(\tau)} = \frac{(1-\alpha)\bar{A}'(\tau)\sigma_i}{(1-\alpha)\bar{A}(\tau)\sigma_i + \rho} = \underbrace{-\frac{1}{\rho}\frac{\partial \gamma}{\partial \tau}}_{\mathrm{MC}} = \frac{1}{\rho}\left(1 - \alpha\bar{A}'(\tau)\right)$$

which holds for all i. We note that $(1-\alpha\bar{A}'(\tau)) > 0$ implies $\tau > \tau^*$. No-one prefers taxes below τ^*.

From Distribution to Growth

Political Economy: Selecting the tax rate



- Illustrating the FOC, and the link to growth. Higher σ_i (capital poorer agent) $\rightarrow$ $MB \uparrow \rightarrow$preferred $\tau \uparrow$.

- Which tax rate will be the one selected through majority voting? On this Alesina and Rodrik invoke the *median voter theorem* (Black, 1948)
  - Preferences are single peaked (unique preferred tax)
  - Uneven (in principle) number of voters
  - $\Rightarrow$ the tax preferred by the median voter is the only one a majority cannot be mobilized against.
- Given the size of $\sigma_M \rightarrow \tau \rightarrow \gamma$ is determined.

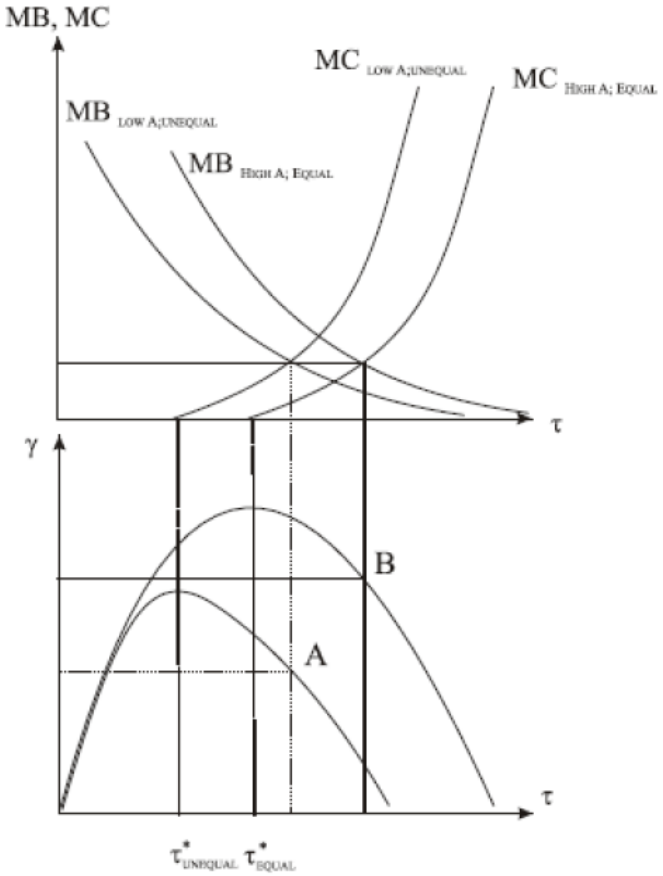
From Distribution to Growth

Predictions, Evidence, and remarks on literature

- **Key prediction**: A (capital) poorer median voter (thus a lower "middelclass share" of wealth), the higher the level of taxation, and the lower the growth rate of the economy.
- Redistribution is quite indirect in this model:
  - Higher taxes allows for greater G, thus higher level of wages and consumption. This gain is relatively greater the greater the importance of wages are to the consumer vis-a-vis wealth.
  - On the other hand: In the long-run higher taxes lower growth of consumption.
  - The less capital you own the more attractive is the intertemporal reallocation of consumption, and thus the higher the level of taxes that you prefer.
- The evicence has not been kind to this mechanism (Perotti, 1996; Milanovic, 2000). All existing studies use cross-sectional data, which might be the problem; the above is conditional on A.

From Distribution to Growth

Predictions, Evidence, and remarks on literature



- Perotti's findings are reconcilable with theory, if A varies and tends to be higher in equal societies.

From Distribution to Growth

Predictions, Evidence, and remarks on literature

- In general inequality will matter to aggregate outcomes, as soon as the a (positive) representative agent ceases to exist (rejecting preferences can be written on Gorman form), or, if you add relevant market imperfections
- Seminar contribution: Galor and Zeira (1988), on the role of credit market imperfections. Means some agents cannot invest in productive capital, implying distribution matters to the macroeconomy.
- A range of approaches has emerged, e.g. linking inequality to political unrest (Alesina and Perotti, 1996), institutional development (Engerman and Sokoloff, 2000), Fertility (De La Croix, & Doepke, 2003), and perhaps the most classic approach, to savings (richer individuals save more)
- Evidence has been mixed. Probably the best recent paper is Erman and te Kaat (2019), using industry level data to identify the effect on productivity. Inequality is bad for human capital intensive sectors (consistent with Galor and Zeira), but "good" for capital intensive sectors (consistent with classical approach).

Roadmap

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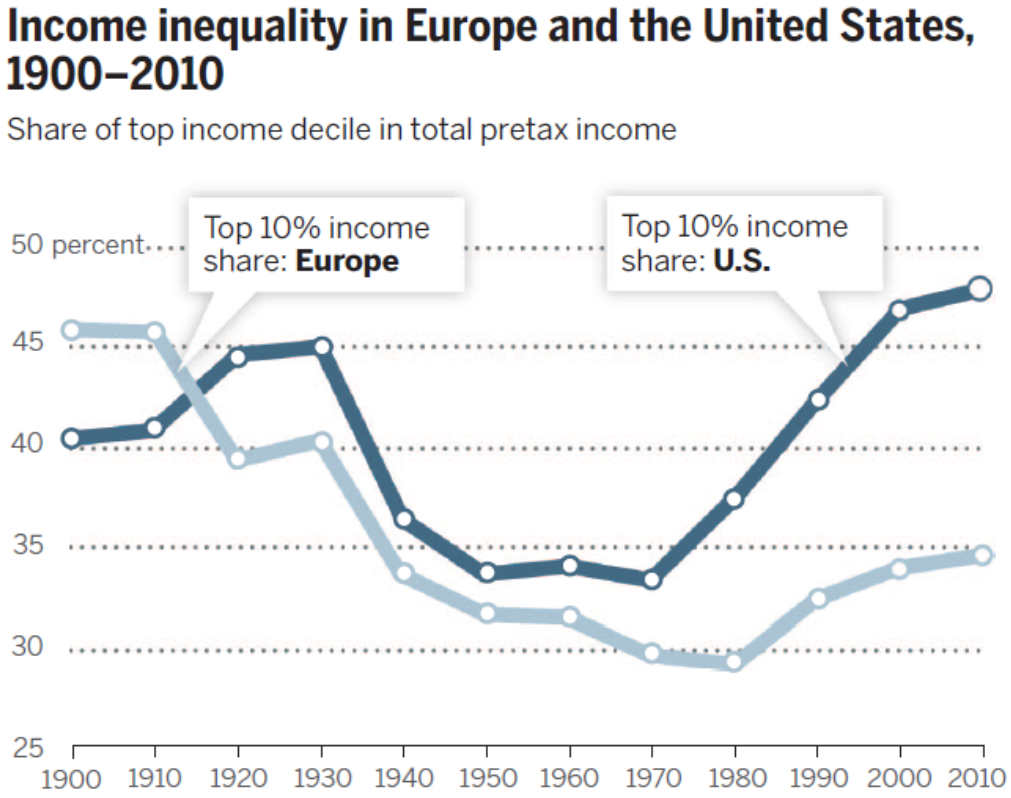
   3. Firm behavior
   4. Political Economy: Selecting taxes
   5. Predictions, Evidence, and remarks on literature
2. **From growth to distribution (Krusell and Smith, Piketty and Saez)**
   1. Data and Piketty's first theory
   2. Summary, critique and remarks on literature

From Growth to Distribution

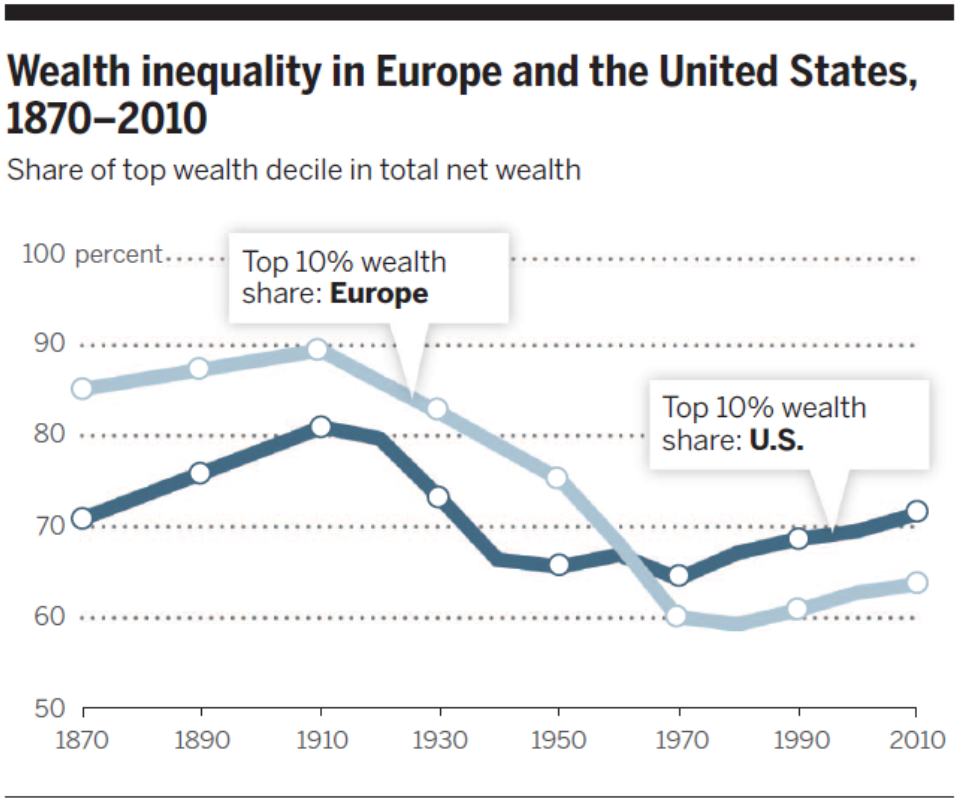
Data



- Dynamics in top inequality over last 100 yrs: U-shaped path

From Growth to Distribution

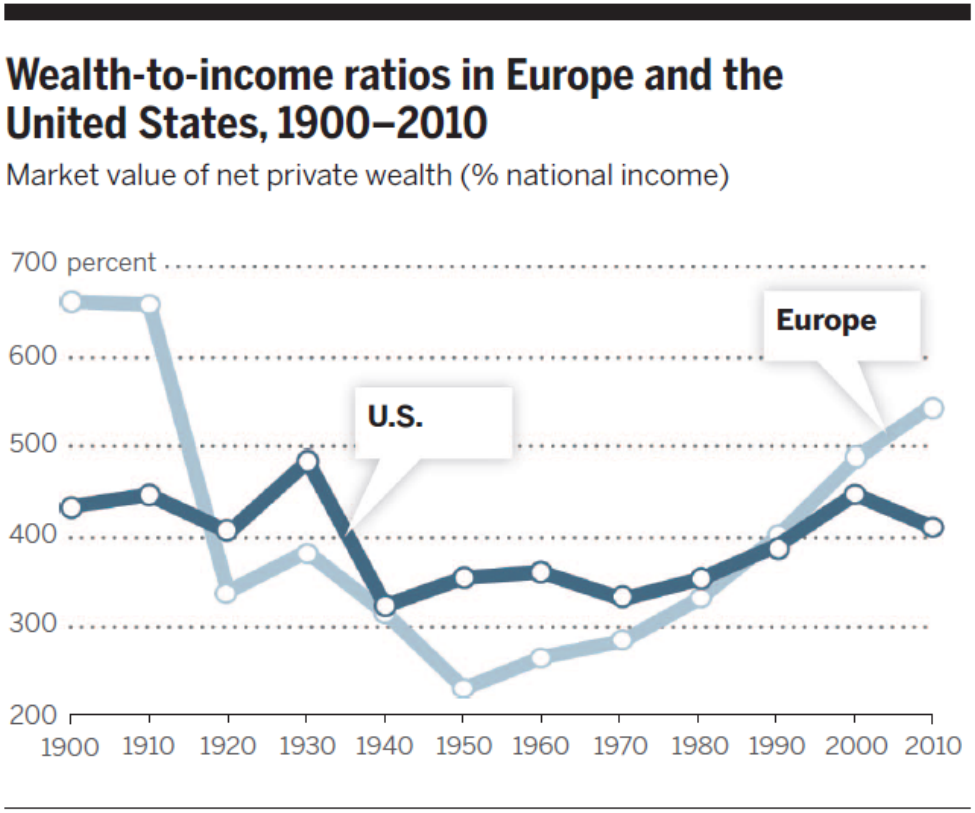
Data



- Wealth inequality: same pattern

From Growth to Distribution

Data



- Capital is "back" (though more pronounced in the US than Europe).

From Growth to Distribution

Piketty's first theory

- Piketty (2014) outlines two theories of (wealth) inequality. The first combines what he calls the first and second law of capitalism:

$$\beta = r\frac{K}{Y} \qquad \text{(1. law)}$$

$$\frac{K}{Y} = \frac{s}{g}, \qquad \text{(2. law)}$$

  where s is net savings and Y is net income (i.e.,both are adjusted for capital depreciation), and g is the growth rate of GDP (technological changes AND population growth, along a steady state path).
- Combining the two:

$$\beta = \frac{rs}{g}.$$

- Basic point: If $g \to 0 \Rightarrow K/Y \to \infty$! May increase inequality, if $\beta \uparrow$ (not a given, however, as discussed below).

From Growth to Distribution

Piketty's first theory

- The first law is basically a definition. The second law, is different. Piketty/Saez calls it the Harrod-Domar-Solow equation.
- But in a standard Solow-Swan model we have (in steady state):

$$\left(\frac{K}{Y}\right)^* = \frac{s}{n+g+\delta},$$

which implies

$$\lim_{n+g\to 0}\left(\frac{K}{Y}\right)^* = \frac{s}{\delta} < \infty. \tag{1}$$

- What is going on?

From Growth to Distribution

Piketty's first theory

- It boils down to which theory of savings you use. Standard Solow-Swan assumption

$$S = \underbrace{s}_{\text{parametrically fixed}} \cdot Y \tag{2}$$

- But we could assume:

$$S - \delta K = \tilde{s}\,(Y - \delta K) \equiv \underbrace{\tilde{s}}_{\text{parametrically fixed}} \cdot \tilde{Y} \tag{3}$$

- Krusell og Smith (KS) calls equation (2) "gross savings", and eqs (3) "net savings".
- Today (2) is the standard assumption. Piketty's modelling choice thus deviates. Implications?

From Growth to Distribution

Piketty's first theory

- In KS notation, Piketty's second law:

$$\frac{K}{\tilde{Y}} = \frac{\tilde{s}}{g}.$$

- Is this plausible? The question KS poses. **Ignore** population growth ("n") from now on, to simplify notation.
- As a point of departure: A Solow-Swan model also holds predictions for capital intensity, with net income replacing gross income:

$$\left(\frac{K}{Y-\delta K}\right)^{*} = \frac{1}{(Y/K)^{*}-\delta} = \frac{1}{(g+\delta)/s-\delta} = \frac{s}{g+\delta(1-s)} \tag{4}$$

- Similarly, the standard Solow-Swan model has predictions regarding $\tilde{s}$, which is NOT fixed ...

From Growth to Distribution

Piketty's first theory

- Using the definition of $\tilde{s}$, we find:

$$\tilde{s} \equiv \frac{S - \delta K}{Y - \delta K} = \frac{\frac{S}{Y} - \delta \frac{K}{Y}}{1 - \delta \left(K / Y \right)} = \frac{s - \delta \left(\frac{K}{Y} \right)^*}{1 - \delta \left(\frac{K}{Y} \right)^*}$$

- Insert steady state solution for $\left(K / Y \right)^*$. You get (after rearrangements):

$$\tilde{s} = \frac{sg}{g + \delta \left(1 - s \right)} \tag{5}$$

- Finally, if we use this equation in expresion for $K / \tilde{Y}$ (eqs 4):

$$\left(\frac{K}{\tilde{Y}} \right)^* = \frac{s}{g + \delta \left(1 - s \right)} = \frac{\tilde{s}}{g}.$$

From Growth to Distribution

Piketty's first theory

- So we can recover Piketty's second law in a Solow-Swan model (steady state)
- BUT, in a standard Solow-Swan model $g \longrightarrow 0$ does not entail K/Y$\longrightarrow \infty$, since $\tilde{s}$ is NOT constant, *cf* eqs *(5)*:

$$\lim_{g \to 0} \frac{\tilde{s}}{g} = \lim_{g \to 0} \frac{sg}{[g + \delta(1-s)]\, g} = \text{“}\frac{0}{0}\text{”}.$$

  L'Hopital gives the correct limit, which is finite (cf equation 1).
- Now, suppose we adopt the Piketty assumption (constant $\tilde{s}$). In this version of the Solow-Swan model, the gross savings rate, s, becomes endogenous...

From Growth to Distribution

Piketty's first theory

- To see how it works, note that:

$$\dot{k} = \tilde{s}\left(y - \delta k\right) \overset{\text{st.}}{\Rightarrow}{}^{\text{st}} g = \tilde{s}\left(\frac{y - \delta k}{k}\right)^{*} \Leftrightarrow \left(\frac{K}{\tilde{Y}}\right)^{*} = \frac{\tilde{s}}{g}$$

  Here $\lim_{g \to 0} \frac{\tilde{s}}{g} = \infty$ as $\tilde{s}$ is constant.
- What about gross savings? We have

$$\frac{S}{Y} = \frac{\tilde{s}\left(y - \delta k\right) + \delta k}{y} = \tilde{s} + \left(1 - \tilde{s}\right)\delta \frac{k}{y}$$

- So if K/Y changes, S/Y changes endogenously. Let's figure out what it is ...

From Growth to Distribution

Piketty's first theory

- First, we need K/Y in this version of the Solow-Swan model. In the steady state we have:

$$gk^* = \tilde{s}\left(y^* - \delta k^*\right) \Leftrightarrow \left(\frac{k}{y}\right)^* = \frac{\tilde{s}}{g + \tilde{s}\delta}$$

- If we then insert $(K/Y)^*$ in the expression for S/Y we get:

$$s = \frac{\tilde{s}\left(g + \delta\right)}{g + \tilde{s}\delta},$$

  which is just (5) "inverted".
- The point: Whether you use s or $\tilde{s}$ as constant *you get exactly the same steady state formulae*
- **BUT: When parameters change predictions will differ**! Different variables are adjusting.

TABLE 1
QUANTITATIVE IMPLICATIONS OF THE MODELS

		Gross Model		Net Model		Net/Gross	
δ	g	k/y	$k/\tilde{y}$	k/y	$k/\tilde{y}$	k/y	$k/\tilde{y}$
.032	.026	3.35	3.75	3.35	3.75	1.00	1.00
.032	.013	4.31	5.00	6.04	7.49	1.40	1.50
.032	.000	6.06	7.52	31.25	∞	5.16	∞
.064	.026	3.35	4.24	3.35	4.24	1.00	1.00
.064	.013	3.90	5.19	5.50	8.48	1.41	1.63
.064	.000	4.69	6.70	15.63	∞	3.33	∞

- Exercise: For δ, g og K/Y given, a value of s is implied. If $\delta = 0.032$, $g = 0.026$ og $K/Y = 3.35$, you get $3.35 = \frac{s}{.026+0.032} \Rightarrow s = 0.1943$. Then g change, and results in "gross-model" is calculated. In net model you choose $\tilde{s}$. Same st st expressions – same results **in baseline**.

From Growth to Distribution

Piketty's first theory

- So which is more plausible? KS note from

$$s = \frac{\tilde{s}\left(g+\delta\right)}{g+\tilde{s}\delta}$$

  that if $g \longrightarrow 0 \Rightarrow s \longrightarrow 1$!
- Interpretation: If $\tilde{s}$ is constant in a stationary economy(no growth), $\dot{k} = \tilde{s}\left(y - \delta k\right) \Rightarrow y^* = \delta k^*$. So net output is zero, and all of gross output is used to maintain the capital stock. Consumption is *zero*.
- Seems a bit agressive. Consistent with utility maximization?

From Growth to Distribution

Piketty's first theory

- Consider "Dynastic consumers" (perfect bequest motive, cf lecture 1). Imagine the economy is stationary ($\dot{c} = 0$, $n = 0$), and preferences are logarithmic($u(c) = \ln(c)$). Consumption Euler implies

$$\dot{c}/c = r - \rho \Rightarrow r = \rho.$$

- Since wealth per capita is constant $\dot{a} = 0$, it must be true that.

$$c = w + ra$$

- Permanent income hypothesis: MPC out of permanet income (wages, return on wealth) = 1.

From Growth to Distribution

Piketty's first theory

- Suppose factor markets are competitive, so that $r = f'(k) - \delta$, and wages reflect marginal product. Also: $a = k$. Then

$$w + ra = \underbrace{f(k) - f'(k)\,k}_{w} + \left(f'(k) - \delta\right)k = f(k) - \delta k \equiv \tilde{y} \Rightarrow c = \tilde{y}$$

- Hence, if we then assume, with Piketty, that

$$c = (1 - \tilde{s})\,\tilde{y}$$

then "Ramsey behavior" requires, $\tilde{s} = 0$! Inconsistent with *ad hoc* assumption $\tilde{s} > 0$.

From Growth to Distribution

Piketty's first theory

- If we instead require (standard in Solow-Swan)

$$c = (1 - s)\, y$$

this is consistent with Ramsey-behavior, where the savings rate then is

$$\dot{k} = 0 \Rightarrow \frac{y - c}{y} \equiv s = \delta \frac{k}{y} = \text{konstant.}$$

- Hence, whereas the standard assumption is reconcilable with Ramsey-behavior, Piketty's assumption is not (in the interesting limit $g \longrightarrow 0$). (In fact, you can choose parameters such that the RCK model produces the same path for capital and consumption as a Solow-Swan model.)
- KS also discuss positive properties: Correlation between s, s and GDP growth. Neither of the models captures the data fully. See paper for details as well as Alvarez-Cuadrado (2019) (See if you can find KS' mistake before reading A-C).

- **Bottom line**:
  - The drammatic effect of $g \to 0$ on K/Y due to "new" savings-behavior.
  - The new specification is not consistent with "Ramsey-behavior". Holds extreme predictions ($s = 1$ in a stationary environment)
  - The key thing is that KS argue their argument is enough to say: Its not a "law". Piketty's theory is not *generally* true. It is not "fundamental".
- Note, as an aside: Good reason to contemplate "$g \to 0$". Demographics: falling n. Productivity? Debate, which we return to later on. But there are those who argue productivity growth also will be declining.

From Growth to Distribution

Piketty's first theory: other issues

- Further issues:
  - If K/Y rises it does not generally lead to an increasing capital *share* (β). Depends on the elasticity of substitution (E) between K and L. Requires E>1. Existing evidence does not uniformly support this.
    - If the share does not increase, rising wealth inequality does not nessesarily translate into income inequality.
  - Actually, it can be argued that K/Y $\uparrow \rightarrow \beta \uparrow$ is less likely if we focus on *net* production and *net* capital share as it affects the size of the elasticity of substitution(see Rognlie, 2016).

From Growth to Distribution

Piketty's Second theory

- Piketty's 2nd teory concerns what he calls the third law of capitalism

$$r > g, \tag{3. law}$$

  where g is GDP growth.
- Basic intuition: In a stationary economy with positive r, wealth *can* outgrow wage income, which suggests greater inequality. (more generally, affects the shape of the wealth distribution and thus *top* share of wealth)
- We return to the inequality above in the context of lecture 3.

From Growth to Distribution

Remarks on literature

- Assuming existence of *positive* representative agent:
- Stiglitz (1969) is among the first to explore the distributional implications of the Solow-Swan model
  - Even if average savings rate is increasing in income, inequality may decline in the long run conditional on structural charactaristics (actually early convergence result)
  - In transition to the steady state: Kuzenet's curve
- Analysis has been extended to the Ramsey-Cass-Koopmans framework (Chatterjee, 1994; Caselli and Ventura, 2000).
  - Also able to produce interesting transitional dynamics of the distribution.

Learning objectives

- The Alesina-Rodrick model from start to finish, eliciting its testable implications.
- The key regularities regarding evolution of capital and wealth, and income inequality over the last century
- Piketty's theories first and second theory of rising inequality.
- The key behavioral assumptions behind the first theory (net savings). The key critique forwarded by KS.
  - Ability to deal with a Solow-Swan model with gross *and* net savings. Derivation of the insight that the two models lead to the same steady state equations in reduced form.
  - Ability to show that the two versions of the model have very different implications when parameters change (i.e., g).

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